

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY,  
KUMASI  
COLLEGE OF SCIENCE  
DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE**



**Predictive Analytics for Kidney Disease Progression:  
A Statistical Examination of Clinical Data**

A THESIS SUBMITTED TO THE DEPARTMENT OF STATISTICS AND ACTUARIAL  
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KUMASI, IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE

OF  
BACHELOR OF SCIENCE  
(STATISTICS)

**February 7, 2025**

## Declaration

We hereby declare that this study, titled "Predicting Chronic Kidney Disease Progression using Predictive Analytics," is our original work and has not been submitted elsewhere for publication. We confirm that we have adhered to ethical standards and properly cited all sources used in this research.

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## Dedication

We dedicate this study to individuals affected by chronic kidney disease, their families, and healthcare professionals dedicated to improving patient outcomes. May this research contribute to enhancing our understanding of CKD progression and inform personalized treatment strategies.

# Abstract

Chronic Kidney Disease (CKD) is a growing health concern, necessitating accurate predictive models for early diagnosis and effective management. This study employs logistic regression to investigate the relationships between key predictor variables and CKD risk. Our analysis reveals that hemoglobin, serum creatinine, hypertension, and random blood glucose are significant predictors of CKD. The developed model demonstrates strong performance, achieving an overall accuracy of 97% and a Kappa value of 0.94 through 10-fold cross-validation. Our findings underscore the importance of these variables in CKD prediction, offering valuable insights for healthcare professionals to enhance disease management and patient outcomes.

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## List of Abbreviations

|             |       |                                      |
|-------------|-------|--------------------------------------|
| <b>CKD</b>  | ..... | Chronic kidney disease               |
| <b>UCI</b>  | ..... | University of California, Irvine     |
| <b>BGR</b>  | ..... | Blood Glucose Random                 |
| <b>ESRD</b> | ..... | End-Stage Renal Disease              |
| <b>GFR</b>  | ..... | Glomerular Filtration Rate           |
| <b>SVMs</b> | ..... | Support Vector Machines              |
| <b>CNNs</b> | ..... | Convolutional Neural Networks        |
| <b>RNNs</b> | ..... | Recurrent Neural Networks            |
| <b>EHRs</b> | ..... | Electronic Health Records            |
| <b>eGFR</b> | ..... | Estimated Glomerular Filtration Rate |

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# Chapter 1

## INTRODUCTION

### 1.1 Background of the study

Chronic kidney disease (CKD) is a widespread health concern with significant implications for public health. In Africa and sub-Saharan Africa, the prevalence rates of CKD are reported as 15.8% and 13.9%, respectively (Stanifer et al. (2014); Kaze et al. (2018)). Specifically in Ghana, the prevalence of CKD is documented at 13.3%, with chronic glomerulonephritis, diabetes mellitus, and hypertension identified as major contributing factors (Adjei et al. (2018); Tannor et al. (2019)). CKD is characterized by the gradual deterioration of kidney function over time, leading to various complications and often necessitating renal replacement therapies like dialysis or transplantation. The kidneys, comprising millions of tiny blood vessels, function as filters to eliminate waste products from the blood (Martinez et al. (2020)). Timely intervention and management are imperative in slowing CKD progression and enhancing patient outcomes.

The conventional approach to managing CKD involves monitoring patients through regular clinical assessments and laboratory tests. However, predicting the trajectory of kidney disease progression remains a complex challenge, as it is influenced by numerous factors, including demographic variables, comorbidities, and lifestyle factors.

In recent years, there has been a growing interest in leveraging predictive analytics and statistical methods to enhance our understanding of CKD progression. These techniques enable a more data-driven and personalized approach to patient care. By analyzing large datasets of clinical information, including patient demographics, laboratory results, and medical history, predictive analytics aims to identify patterns and correlations that can

inform prognostic models.

The integration of statistical methods in the analysis of clinical data offers a systematic way to extract meaningful insights. These methods can help identify risk factors, establish predictive models, and contribute to the development of more accurate and efficient tools for assessing kidney disease progression. The ultimate goal is to enable healthcare professionals to intervene earlier, tailor treatment plans, and improve the overall management of CKD.

As technology advances and healthcare systems increasingly embrace data-driven approaches, the application of predictive analytics to kidney disease holds great promise. This study seeks to contribute to this evolving field by conducting a detailed statistical examination of clinical data, aiming to uncover valuable insights that can advance our ability to predict and manage kidney disease progression effectively. Through the integration of predictive analytics, the study aims to pave the way for more proactive and personalized strategies in the care of individuals with CKD.

## 1.2 Problem Statement

Chronic kidney disease (CKD) poses a substantial and escalating health challenge worldwide, characterized by a progressive decline in renal function. Despite advances in medical science, accurately predicting the trajectory of CKD progression remains a complex and often elusive task. Current clinical methods for assessing disease advancement rely on periodic evaluations and static risk factors, lacking the precision needed for timely intervention and personalized treatment strategies.

The inherent variability in CKD progression, influenced by multifaceted factors such as patient demographics, comorbidities, and lifestyle variables, underscores the inadequacy of traditional approaches. Conventional methods often result in delayed identification of high-risk individuals and may not effectively capture the dynamic nature of the disease.

## 1.3 Purpose of the Study

This study recognizes the critical need to address these limitations and seeks to leverage the power of predictive analytics and statistical examination of clinical data. By doing so, the aim is to discern intricate patterns, correlations, and risk factors that elude conventional analysis. The overarching problem lies in the absence of a comprehensive and precise predictive model that integrates diverse clinical data to foresee the progression of CKD accurately.

## 1.4 Objective of the Study

The primary objective of this study is to develop and implement a robust predictive analytics framework for accurately forecasting the progression of chronic kidney disease (CKD) through a comprehensive statistical examination of clinical data. The specific aims include:

- Explore Data Characteristics and Prepare for Analysis
- Select and Train Predictive Models
- Evaluate Model Performance
- Optimize Model Performance

## 1.5 Research Questions

Specific research questions related to the objectives, include:

- What are the primary clinical factors that correlate with the progression of kidney disease?

- How accurately can predictive models forecast disease progression using secondary data?

## 1.6 Significance of the Study

Chronic kidney disease (CKD) represents a significant burden on global healthcare systems, leading to substantial morbidity and mortality. Early detection and effective management of CKD are crucial for improving patient outcomes and reducing healthcare costs. By developing a robust predictive analytics framework for forecasting CKD progression, this study aims to address several critical issues:

- **Early Intervention:** Early identification of individuals at risk of CKD progression allows for timely intervention strategies, including lifestyle modifications and pharmacological interventions, which can potentially slow down disease progression and prevent complications.
- **Healthcare Resource Allocation:** Accurate prediction of CKD progression facilitates better allocation of healthcare resources by identifying high-risk patients who may require more intensive monitoring and interventions, thereby optimizing healthcare delivery and reducing costs.
- **Research and Development:** Insights gained from predictive analytics can inform future research directions, guiding the development of novel diagnostic tools, therapeutic interventions, and healthcare policies aimed at improving CKD management and outcomes.
- **Public Health Implications:** By improving our understanding of CKD progression and risk factors, this study has broader public health implications, potentially informing preventive strategies and population-level interventions aimed at reducing the incidence and burden of CKD.

## 1.7 Scope and the Delimitation of the Study

This study focuses on developing and implementing a predictive analytics framework for forecasting the progression of chronic kidney disease (CKD) using clinical data. The scope of the study includes:

- **Data Collection:** The study will utilize a dataset containing clinical information of individuals with and without CKD, focusing on variables such as age, blood pressure, laboratory test results, and comorbidities.
- **Predictive Modeling:** The study will involve the exploration of various predictive modeling techniques, including machine learning algorithms, to develop accurate prognostic models for CKD progression.
- **Model Evaluation:** The developed models will be rigorously evaluated using appropriate performance metrics to assess their predictive accuracy, sensitivity, specificity, and generalizability.
- **Optimization Strategies:** Strategies for optimizing model performance, including feature selection, parameter tuning, and ensemble methods, will be explored to enhance the reliability and robustness of the predictive analytics framework.
- **Limitations:** The study acknowledges certain limitations, including the availability and quality of the clinical dataset, potential biases inherent in retrospective data analysis, and the generalizability of findings to diverse populations.

The study does not encompass experimental interventions or clinical trials but rather focuses on secondary data analysis and model development using existing clinical datasets.

## 1.8 Operational Definition of Terms

To ensure clarity and consistency in terminology, the following operational definitions are adopted:

**Chronic Kidney Disease (CKD):** A progressive condition characterized by the gradual loss of kidney function over time, typically defined by abnormalities in kidney structure or function persisting for more than three months.

**Predictive Analytics:** The use of statistical techniques and machine learning algorithms to analyze data and make predictions about future events or outcomes, in this case, predicting the progression of CKD based on clinical variables.

**Prognostic Model:** A mathematical model developed using predictive analytics techniques to estimate the likelihood of future clinical events or outcomes, such as CKD progression, based on patient characteristics and other relevant factors.

**Feature Selection:** The process of selecting a subset of relevant variables or features from a larger set of potential predictors to improve the performance and interpretability of predictive models.

**Machine Learning Algorithms:** Computational algorithms that enable computers to learn from data and make predictions or decisions without being explicitly programmed, including techniques such as decision trees, support vector machines, and neural networks.

**Model Evaluation Metrics:** Quantitative measures used to assess the performance of predictive models, including accuracy, sensitivity, specificity, area under the receiver operating characteristic curve (AUC-ROC), and precision-recall curves.

## Chapter 2

### Literature Review

#### 2.1 Introduction

Chronic Kidney Disease (CKD) is a significant public health challenge, affecting millions worldwide and often leading to end-stage renal disease (ESRD) and increased cardiovascular risk. Accurate prediction of CKD progression is vital for timely intervention, yet traditional statistical models have their limitations. Data analytics and machine learning offer promising avenues to enhance the prediction and personalized risk assessment of CKD progression. This literature review delves into current research on applying data analytics for CKD progression, examining model development, advanced analytical approaches, existing challenges, and potential future directions. Through synthesizing existing evidence, this review aims to highlight the potential of data analytics in improving clinical management and outcomes for CKD patients.

#### 2.2 Conceptual Review

The progression of CKD is a complex process influenced by a variety of demographic, clinical, laboratory, and lifestyle factors. Accurate prediction is essential for effective management and timely intervention.

##### 2.2.1 Risk Factors

Several key risk factors have been identified that contribute to CKD progression:

- **Demographic factors:** Age, sex, race, and socioeconomic status are significant predictors of CKD progression (Tangri et al. (2011)).
- **Clinical history:** Conditions such as diabetes, hypertension, cardiovascular disease, and obesity are critical in determining the risk of CKD progression (Tangri et al. (2011); Segal et al. (2020)).
- **Laboratory measurements:** Indicators like lower glomerular filtration rate (GFR) at diagnosis, proteinuria, anemia, metabolic acidosis, and disorders of bone and mineral metabolism are vital predictors (Tangri et al. (2011); Segal et al. (2020)).
- **Lifestyle factors:** Behaviors such as smoking, physical inactivity, and unhealthy diet significantly impact CKD progression (Tangri et al. (2011); Segal et al. (2020)).

### 2.2.2 Key Predictive Models and Techniques

Various models and techniques have been developed to predict CKD progression:

- **Logistic Regression and Cox Proportional Hazards Models:** These traditional statistical methods are commonly used for predicting CKD progression. Logistic regression is valuable for predicting binary outcomes, such as progression to ESRD within a specified period (Tangri et al. (2017); Segal et al. (2020)). The Cox proportional hazards model estimates the risk of CKD progression over time, considering covariates such as age, gender, blood pressure, and baseline kidney function (Liu et al. (2012); Chang et al. (2021)).
- **Machine Learning Algorithms:**
  - **Random Forests:** These models, which leverage multiple decision trees, have been used to predict CKD progression. They effectively handle complex interactions between variables and manage missing data (Chittora et al. (2021); Segal et al. (2020)).

- **Support Vector Machines (SVMs):** SVMs have shown high accuracy in classifying CKD stages and work well with high-dimensional data common in medical datasets (Chittora et al. (2021); Segal et al. (2020)).
- **Deep Learning Models:** Techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) capture non-linear relationships and temporal patterns in CKD progression data (Liu et al. (2012); Segal et al. (2020)).
- **Advanced Survival Analysis Techniques:** Time-varying covariate models and survival trees provide a nuanced understanding of CKD progression, allowing for dynamic prediction based on time-dependent risk factors (Tangri et al. (2017); Chang et al. (2021)).

### 2.2.3 Limitations and Challenges

Despite advances, several challenges remain in predicting CKD progression:

- **Data Quality and Integration:** Ensuring the quality and consistency of data used in predictive models is a primary challenge. Inconsistent data collection methods, missing values, and variations in measurement units can undermine model accuracy (Segal et al. (2020)). Integrating data from diverse sources, such as electronic health records (EHRs), laboratory results, and patient-reported outcomes, requires sophisticated harmonization techniques to ensure compatibility and reliability (Tangri et al. (2011)).
- **Model Complexity vs. Interpretability:** While advanced models like neural networks offer high accuracy, their "black-box" nature can hinder clinical adoption due to a lack of transparency (Liu et al. (2012)). Explainable AI (XAI) methods are essential to enhance model interpretability, build clinician trust, and facilitate integration into clinical workflows (Chittora et al. (2021)).
- **Generalizability Across Populations:** Predictive models often perform poorly when applied to populations different from those on which they were developed.

External validation across diverse patient cohorts is crucial to ensure the robustness and applicability of these models in real-world settings (Chang et al. (2021); Tangri et al. (2017)).

Addressing these challenges is essential for the successful integration of data analytics in CKD management. High-quality data, model interpretability, and broad validation are key to developing trustworthy predictive tools that can inform clinical decision-making, guide personalized treatment strategies, and ultimately improve patient outcomes.

## 2.3 Empirical Review

This section reviews empirical studies on predictive modeling for CKD progression, highlighting significant advancements and challenges.

### 2.3.1 Overview of Predictive Models

A scoping review by (Lim et al. (2022)) examined 516 studies, of which 33 developed statistical or computational models to predict CKD progression. Most studies employed supervised learning techniques, with Cox regression modeling being the most prevalent. However, heterogeneity in study populations and data sources complicates comparisons across models. The review noted a lack of consistency in reporting model development and validation, which poses challenges for reproducibility and clinical utility.

(Fu et al. (2024)) conducted a study in Singapore, developing a nationwide predictive model to identify high-risk CKD patients in the early stages. This model, using demographic, clinical, and laboratory data, outperformed traditional clinical assessments, underscoring the potential of machine learning in enhancing predictive accuracy and supporting clinical decision-making.

### 2.3.2 Machine Learning Approaches

Machine learning has gained traction in CKD research:

- A study published in *Kidney Medicine* developed a 7-variable risk classifier that accurately predicted a decline in estimated glomerular filtration rate (eGFR) of more than 30% within five years. The model demonstrated a high area under the curve (AUC) for receiver operating characteristic (ROC) analysis, proving effective in risk stratification for CKD progression (Aoki et al. (2023)).
- A systematic review by (McMahon et al. (2021)) highlighted the potential of ML techniques to explore complex, non-linear relationships within large datasets, improving the understanding of CKD progression and informing key interventions.

### 2.3.3 Challenges and Limitations

Challenges persist in predictive modeling for CKD:

- **Lack of Standardized Reporting:** The absence of standardized guidelines complicates the comparison of models and their performance across populations. The Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD) statement has been proposed to improve transparency and reproducibility in predictive modeling studies (McMahon et al. (2021)).
- **Generalizability Issues:** Models developed using specific populations may not generalize well to broader patient populations. External validation is crucial to ensure effective application across varied clinical settings (Tangri et al. (2022)).

### 2.3.4 Conclusion

The empirical review reveals a growing body of research focused on developing predictive models to enhance CKD management. While significant progress has been made, challenges related to data quality, model validation, and reporting standardization remain. Future research should address these challenges to facilitate the broader adoption of predictive analytics in CKD management.

## 2.4 Theoretical Review

The theoretical review explores foundational concepts and methodologies in predictive analytics for CKD progression, aiming to enhance early detection and personalize treatment strategies.

### 2.4.1 Key Concepts

- **Predictive Modeling Techniques:** Traditional statistical methods, such as Cox proportional hazards models and logistic regression, are contrasted with modern machine learning approaches. While traditional models often assume linear relationships and rely on a limited number of variables, machine learning techniques can handle complex datasets and capture non-linear relationships, improving predictive accuracy.
- **Empirical Evidence:** The literature reflects a growing application of predictive models in CKD progression. A systematic review highlighted variability in model development and validation, emphasizing supervised learning techniques. Studies by (Fu et al. (2024)) and (Lim et al. (2022)) demonstrate the effectiveness of predictive models in identifying high-risk patients and improving clinical decision-making.
- **Challenges and Limitations:** Despite advancements, challenges such as inconsistent

reporting methodologies, data quality issues, and integrating predictive models into clinical practice persist. These challenges hinder the reproducibility and clinical utility of predictive analytics.

### **2.4.2 Conclusion**

Predictive analytics holds the potential to transform CKD management through enhanced risk stratification and personalized interventions. However, addressing challenges related to data quality, methodological consistency, and clinical integration is crucial for the successful implementation of predictive models. This review underscores the importance of ongoing research and collaboration among clinicians, data scientists, and policymakers to optimize the use of data analytics in improving CKD patient outcomes.

## Chapter 3

# METHODOLOGY

### 3.1 Introduction

In this chapter, we present the methodology used to examine the diverse factors influencing the progression of kidney diseases. Our objective is to provide a comprehensive overview of the research design, data collection procedures, data analysis methods, as well as the ethical considerations taken into account throughout the study. By elucidating these elements, we ensure a robust and systematic approach towards developing an effective model for predicting kidney disease progression.

### 3.2 Research Design

The research design adopted for this study is a quantitative, observational, and cross-sectional design. This approach allows for the systematic collection and analysis of data to identify patterns and relationships between variables associated with kidney disease progression. The study aims to develop predictive models using machine learning techniques, which necessitates a rigorous and structured methodology.

The key steps in the research design include:

1. **Literature Review:** A comprehensive review of existing literature on kidney disease progression, predictive modeling, and relevant clinical parameters is conducted. This review helps in identifying the key factors influencing kidney disease progression and informs the selection of attributes for the predictive models.

2. **Data Acquisition:** The Chronic Kidney Disease (CKD) dataset from the UCI Machine Learning Repository is selected as the primary data source. This dataset provides a robust foundation for developing and validating the predictive models.
3. **Data Preprocessing:** The acquired data is subjected to preprocessing steps, including handling missing values, normalization, and encoding of categorical attributes. These steps are crucial to ensure the data's suitability for analysis and modeling.
4. **Exploratory Data Analysis (EDA):** EDA is performed to gain insights into the data distribution, identify potential outliers, and understand the relationships between variables. Visualization techniques such as histograms, scatter plots, and correlation matrices are utilized during this phase.
5. **Model Development:** Multiple machine learning algorithms, including logistic regression, decision trees, random forests, support vector machines, and neural networks, are employed to develop predictive models. The models are trained on the training dataset, and hyperparameter tuning is performed to optimize their performance.
6. **Model Evaluation:** The performance of the predictive models is evaluated using the testing dataset. Evaluation metrics such as accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve are calculated to assess the models' effectiveness.
7. **Validation and Testing:** Cross-validation techniques are applied to ensure the robustness and generalizability of the predictive models. The models are also tested on an independent validation dataset, if available, to further verify their performance.
8. **Interpretation and Reporting:** The results of the data analysis and model evaluation are interpreted in the context of existing literature and clinical knowledge. The findings are documented and reported in a structured format, highlighting the implications for clinical practice and future research.

The research design ensures a methodical approach to developing and validating predictive models for kidney disease progression, leveraging the strengths of machine learning techniques and robust data analysis methods.

### 3.3 Selection of Secondary Data Set

In this study, the Chronic Kidney Disease (CKD) dataset provided by Apollo Hospitals, India, is utilized as the primary source of secondary data. This data set is obtained from the UCI Machine Learning Repository and consists of valuable clinical information collected from individuals in the early stages of CKD (Kerr et al. (2012)).

The CKD data set is contributed by Dr. P. Soundarapandian, a Senior Consultant Nephrologist at Apollo Hospitals, India. The data set was curated by L. Jerlin Rubini, a Research Scholar at Alagappa University, under the guidance of Dr. P. Eswaran, an Assistant Professor in the Department of Computer Science and Engineering at Alagappa University (Rubini and Eswaran (2015)).

This data set comprises 400 instances, with 250 instances representing individuals diagnosed with CKD and 150 instances representing individuals without CKD. It contains a total of 25 attributes, including 11 numerical and 14 nominal attributes. These attributes encompass various clinical parameters such as age, blood pressure, specific gravity, albumin, sugar, and several others, along with the target class attribute indicating the presence or absence of CKD.

Given the comprehensive nature of the CKD data set and its relevance to our research objectives, it serves as an ideal secondary data source for our study on predicting CKD progression using predictive analytics methods.

### 3.4 Data Preparation and Cleaning

The quality and integrity of the data set are paramount for the success of our analysis. Therefore, meticulous data preparation and cleaning procedures are implemented to ensure that the data is suitable for predictive modeling.

We begin by examining the data set for missing values, outliers, and inconsistencies. Missing values are addressed through imputation techniques, such as mean, median, or mode imputation, depending on the nature of the missing data and the distribution of variables.

**Mean Imputation:** Mean imputation involves replacing missing values with the mean of the non-missing values in the respective column. Given a dataset  $X$  with a column  $x$  containing missing values  $x_{\text{missing}}$ , the mean  $\bar{x}$  is calculated as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where  $n$  is the number of non-missing values. The missing values  $x_{\text{missing}}$  are then replaced with  $\bar{x}$ . Mean imputation is suitable when the data distribution is approximately normal and missing values are assumed to be missing at random (MAR).

**Median Imputation:** Median imputation involves replacing missing values with the median of the non-missing values in the respective column. Given a dataset  $X$  with a column  $x$  containing missing values  $x_{\text{missing}}$ , the median  $\tilde{x}$  is calculated as:

$$\tilde{x} = \text{median}(x)$$

The missing values  $x_{\text{missing}}$  are then replaced with  $\tilde{x}$ . Median imputation is robust to outliers and is suitable when the data contains extreme values or is skewed.

**Mode Imputation:** Mode imputation involves replacing missing values with the mode (most frequent value) of the non-missing values in the respective column. Given a dataset  $X$  with a column  $x$  containing missing values  $x_{\text{missing}}$ , the mode  $\text{Mode}(x)$  is calculated as the most frequent value. The missing values  $x_{\text{missing}}$  are then replaced with  $\text{Mode}(x)$ . Mode imputation is used for categorical variables.

Furthermore, variables with more than 30% missing values are removed from the data set, as retaining such variables may introduce bias and reduce the effectiveness of the analysis (Aljaaf (2017)). This decision is supported by the need to maintain the integrity and reliability of the data set, ensuring that the resulting predictive models are built on high-quality data.

Next, categorical variables are converted into factors, and levels are assigned to them to facilitate the analysis and modeling process. This conversion process involves assigning a numerical representation to each category within the categorical variables, enabling the algorithms to interpret and utilize these variables effectively during analysis.

Furthermore, redundant or irrelevant variables identified during the correlation analysis are removed to streamline the data set and improve the efficiency of the modeling process.

## 3.5 Data Analysis Techniques

To fulfill the objectives of our study, we employ predictive analytics methods aimed at examining the relevance of input independent variables to the dependent variable (class attribute) in predicting chronic kidney disease (CKD) progression. Our focus lies in understanding both the significance of individual independent variables and the associations between them.

### 3.5.1 Relationships Between Variables

We begin by measuring the correlation among independent variables using correlation matrix. This matrix is computed by first obtaining the variance covariance matrix  $\Sigma$  of the  $p$  numeric variables.

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2p} \\ \vdots & \vdots & \dots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \dots & \sigma_{pp} \end{bmatrix}$$

Since  $\sigma_{jk} = \sigma_{kj}$  for all  $j \neq k$  we note that  $\Sigma$  is symmetric and is also non-negative definite matrix.

The  $p \times p$  standard deviation matrix  $V^{1/2}$  is a diagonal matrix with  $\sqrt{\sigma_{jj}}$  along the diagonal and zeros in all off-diagonal positions. Then

$$\Sigma = V^{1/2} P V^{1/2}$$

and the correlation matrix is

$$(V^{1/2})^{-1} \Sigma (V^{1/2})^{-1}$$

Given  $\Sigma$ , we can easily obtain the correlation matrix

$$(V^{1/2})^{-1} \Sigma (V^{1/2})^{-1} = \begin{bmatrix} 1 & \rho_{12} & \dots & \rho_{1p} \\ \rho_{21} & 1 & \dots & \rho_{2p} \\ \vdots & \vdots & \dots & \vdots \\ \rho_{p1} & \rho_{p2} & \dots & 1 \end{bmatrix}$$

The correlation matrix is the  $p \times p$  matrix with off-diagonal elements equal to  $\rho_{jk}$  and diagonal elements equal to 1 . Since  $\rho_{ij} = \rho_{ji}$  the correlation matrix is symmetric and non-negative definite. This analysis enables us to assess the strength of relationships between the numeric predictors, allowing us to identify and address any issues of multicollinearity.

The Chi-Square test for association is used to determine if there is a significant association between two categorical variables.

The test statistic is calculated as:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

- $O_{ij}$  is the observed frequency in the  $i$ -th row and  $j$ -th column,
- $E_{ij}$  is the expected frequency in the  $i$ -th row and  $j$ -th column,
- $r$  is the number of rows,
- $c$  is the number of columns.

This test enabled us to identify whether the observed patterns in the data are due to chance or a genuine relationship between the variables.

### 3.5.2 Dimensionality Reduction

Principal Component Analysis(PCA) is a dimensionality reduction technique that helped identify the most important variables that explain the maximum variance in the data set.

We begin by obtaining the variance covariance matrix  $\Sigma$  of the  $p$  numeric variables.

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2p} \\ \vdots & \vdots & \dots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \dots & \sigma_{pp} \end{bmatrix}$$

We obtain the eigenvalues and eigenvectors of  $\Sigma$  and arrange them in descending order.

$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \dots \geq \lambda_p \geq 0$  are the eigenvalues of  $\Sigma$

$e_1, e_2, e_3, e_4 \dots e_p$  are the corresponding eigenvectors of  $\Sigma$

Vector of measurements on the  $j$ -th observation

$$X_j = \begin{bmatrix} X_{1j} \\ X_{2j} \\ \vdots \\ X_{pj} \end{bmatrix}, j = 1, \dots, n$$

Suppose  $x_j \sim N(\mu, \Sigma)$ ,

First Principal Component:

$$Y_{1j} = e_1' X_j = e_{11}X_{1j} + e_{21}X_{2j} + \dots + e_{p1}X_{pj}$$

Moments:

$$E(Y_{1j}) = e_1' \mu = \sum_{i=1}^p e_{i1} \mu_i$$

$$\text{Var}(Y_{1j}) = \text{Var}(e_1' X_j) = e_1' \Sigma e_1 = e_1' (\lambda_1 e_1) = \lambda_1$$

Second Principal Component:

$$Y_{2j} = e_2' X_j = e_{12}X_{1j} + e_{22}X_{2j} + \dots + e_{p2}X_{pj} = \sum_{i=1}^p e_{i2}X_{ij}$$

and

$$E(Y_{2j}) = e_2' \mu = \sum_{i=1}^p e_{i2} \mu_i$$

$$\text{Var}(Y_{2j}) = \text{Var}(e_2' X_j) = e_2' \Sigma e_2 = \lambda_2$$

$$\text{cov}(Y_{1j}, Y_{2j}) = \text{cov}(e_1' X_j, e_2' X_j) = e_1' \Sigma e_2 = e_1' (\lambda_2 e_2) = 0$$

k-th Principal Component:

$$Y_{kj} = e'_k x_j = \sum_{i=1}^p e_{ik} X_{ij}$$

and

$$E(Y_{kj}) = e'_k \mu$$

$$\text{Var}(Y_{kj}) = \text{Var}(e'_k X_j) = e'_k \Sigma e_k = \lambda_k$$

$$\text{cov}(Y_{kj}, Y_{rj}) = \text{cov}(e'_k X_j, e'_r X_j) = e'_k \Sigma e_r = e'_k (\lambda_r e_r) = 0$$

Total Variance:

$$\begin{aligned} \text{trace}(\Sigma) &= \text{Var}(X_{1j}) + \cdots + \text{Var}(X_{pj}) \\ &= \lambda_1 + \cdots + \lambda_p \\ &= \text{Var}(Y_{1j}) + \cdots + \text{Var}(Y_{pj}) \end{aligned}$$

Proportion of total variance accounted for by the k -th principal component

$$\frac{\text{Var}(Y_k)}{\sum_{i=1}^{\mu} \text{Var}(Y_{ij})} = \frac{\lambda_k}{\sum_{i=1}^{\nu} \lambda_i}$$

Proportion of the total variance accounted for by the first  $k$  principal components:

$$\frac{\sum_{i=1}^k \text{Var}(Y_{ij})}{\sum_{i=1}^{\mu} \text{Var}(Y_{ij})} = \frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^{\mu} \lambda_i}$$

When the first  $k$  principal components explain a significant portion of the total variance, we focus on the corresponding variables,  $Y_j, \dots, Y_{kj}$ . To identify the most influential variables, we examine the absolute values of the coefficients in the eigenvectors of these top  $k$  components. The variables with the largest absolute coefficients in these components are deemed the most important, as they contribute the most to the variance explained by the principal components.

### 3.5.3 Model

Once we have identified the most significant variables, we employ machine learning algorithms, specifically logistic regression, to predict the progression of kidney disease. This algorithm is chosen because the response variable is dichotomous.  $Bernoulli(\pi)$  random variable takes values zero and one with probabilities  $1 - \pi$  and  $\pi$ , respectively. These random variables arise because individuals are classified into two categories, **ckd** and **notckd**, say, and we define

$$Y = \begin{cases} 1 & \text{if the individual is a ckd patient} \\ 0 & \text{if the individual is notckd patient} \end{cases}$$

with

$$P(Y = 1) = \pi$$

$$P(Y = 0) = 1 - \pi$$

Suppose  $Y_1, Y_2, \dots, Y_n$  are independent Bernoulli ( $\pi_i$ ) random variables, so

$$E(Y_i) = P(Y_i = 1) = \pi_i.$$

The logistic regression model is

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 X_i$$

which can be written as

$$\pi_i = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}}$$

where for  $i = 1, 2, \dots, n$

$X_i$  is the value of the predictor variable  $X$  for the  $i$  th individual

$\pi_i = P(Y_i = 1)$  is the probability that the  $i$  th individual's response is one and  $\beta_0$  and  $\beta_1$  are parameters of the model.

The coefficients  $\beta_0, \beta_1, \dots, \beta_k$  are estimated using maximum likelihood estimation (MLE).

$$P(Y|X, \beta) = \prod_{i=1}^n \left( \frac{e^{\beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik}}}{1 + e^{\beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik}}} \right)^{y_i} \left( \frac{1}{1 + e^{\beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik}}} \right)^{1-y_i}$$

Where:

$y_i$  = Observed outcome for individual  $i$

$X_{ij}$  = Value of predictor  $j$  for individual  $i$

yields the log likelihood:

$$\ell(\beta) = \sum_{i=1}^n [y_i (\beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik}) - \log (1 + e^{\beta_0 + \beta_1 X_{i1} + \dots + \beta_k X_{ik}})]$$

Maximizing the likelihood (or log likelihood) has no closed-form solution, so a technique like iteratively reweighted least squares is used to find an estimate of the regression coefficients,  $\hat{\beta}$ .

Once the maximum likelihood estimates  $b_0$  and  $b_1$  of  $\beta_0$  and  $\beta_1$  are obtained, the fitted values, denoted  $\tilde{\pi}_1, \tilde{\pi}_2, \dots, \tilde{\pi}_i$ , are defined as

$$\tilde{\pi}_i = \frac{e^{b_0 + b_1 x_i}}{1 + e^{b_0 + b_1 x_i}} \quad \text{for } i = 1, 2, \dots, 2$$

The odds of a patient classified as ckd ( $Y = 1$ ) is

$$\frac{\pi}{1 - \pi} = \frac{P(Y = 1)}{P(Y = 0)}$$

The odds ratio of an individual's response being classified as ckd ( $Y = 1$ ), for a one-unit increase in  $X$ , is

$$\tilde{OR} = \frac{\tilde{\pi}_2/(1 - \tilde{\pi}_2)}{\tilde{\pi}_1/(1 - \tilde{\pi}_1)}$$

With the model defined and estimated, we now turn our attention to evaluating its performance and validity.

### 3.5.4 Model Validation

Next, we train the selected model on the prepared dataset using a technique Cross-Validation. This approach ensures that our model evaluations are reliable, robust, and generalizable, reducing the risk of overfitting and improving the model's ability to perform well on unseen data. Cross-Validation involves dividing the dataset into complementary subsets, typically into  $k$  folds, to create multiple training and testing sets. Mathematically, let  $D$  denote the data set,  $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ , where  $x_i$  represents the features and  $y_i$  represents the labels. We partition  $D$  into  $k$  disjoint subsets  $D_1, D_2, \dots, D_k$ . Then, for each  $i$  from 1 to  $k$ , we train the model on  $D - D_i$  and evaluate it on  $D_i$ .

By using Cross-Validation, we can create a more robust and reliable model evaluation framework that minimizes the risk of overfitting and ensures our models are generalizable to new data.

### 3.5.5 Model Performance

Following model training, we assess the performance of the trained models using evaluation metrics such as,

- **Accuracy:** Calculate the accuracy of the model by comparing the predicted labels with the true labels and calculating the proportion of correctly classified instances.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

- **Precision:** Precision measures the proportion of true positive predictions among all positive predictions made by the model. It focuses on the correctness of positive predictions.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- **Recall:** Recall measures the proportion of true positive predictions among all actual positive instances in the data set. It focuses on the model's ability to identify all positive instances.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

- **F1-score:** The F1-score is the harmonic mean of precision and recall. It provides a balance between precision and recall and is useful when there is an imbalance between classes.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

## 3.6 Summary

To achieve the research objectives, we investigate kidney disease progression using predictive analytics with the CKD data set from the UCI Machine Learning Repository. Data preparation involves handling missing values and converting variables, followed by correlation analysis and statistical tests to understand variable relationships. Logistic regression is utilized for prediction, with evaluation metrics such as accuracy, precision, recall, and F1-score used to assess model performance, while cross-validation techniques ensure robustness.

## Chapter 4

# ANALYSIS, RESULTS AND DISCUSSION

### 4.1 Introduction

This chapter presents the findings from the analysis of the Chronic Kidney Disease (CKD) data set obtained from Apollo Hospitals, India, and explores the factors influencing CKD progression using predictive analytics methods. The CKD data set, sourced from the UCI Machine Learning Repository, comprises 400 instances with 25 attributes, including clinical parameters such as age, blood pressure, specific gravity, albumin, and sugar levels, along with a target class attribute indicating the presence or absence of CKD. Out of the 400 instances, 250 represent individuals diagnosed with CKD, while 150 represent individuals without CKD.

Our analysis begins with a thorough data preparation and cleaning process, addressing missing values through imputation techniques and converting categorical variables into factors. Descriptive statistics, correlation analysis and statistical tests, such as Pearson's correlation coefficient, Chi-Square test, and ANOVA, were employed to understand the relationships between variables and their impact on CKD progression.

Following data preparation, we employ logistic regression to predict CKD progression. The model's performance is evaluated using metrics like accuracy, precision, recall, and F1-score, with cross-validation techniques ensuring robustness and mitigating the risk of overfitting.

In addition to presenting the results, this chapter discusses the implications of our findings in the context of existing literature and clinical practices. We aim to identify significant predictors of CKD progression and assess the practical applicability of the predictive models

developed.

By thoroughly examining the results and their implications, this chapter aims to provide a deeper understanding of CKD progression and contribute valuable insights to the field of healthcare analytics.

## 4.2 Descriptive Statistics

Table 4.1 presents the descriptive statistics for the numeric variables in our data set. It provides an overview of the central tendency and variability of each numeric variable in our data set. We can see that the means and medians of the variables vary widely, indicating differences in the distribution of each variable. Additionally, the standard deviations and ranges (Min-Max) provide insight into the dispersion and range of values for each variable.

### 4.2.1 Numeric Variables

Table 4.1: Descriptive Statistics for Numeric Variables

| Variable               | Mean    | Std Dev | Min     | Q1      | Median  | Q3      | Max      |
|------------------------|---------|---------|---------|---------|---------|---------|----------|
| Age                    | 51.56   | 16.98   | 2.00    | 42.00   | 55.00   | 64.00   | 90.00    |
| Blood Glucose Random   | 145.06  | 75.26   | 22.00   | 101.00  | 121.00  | 150.00  | 490.00   |
| Blood Pressure         | 76.58   | 13.49   | 50.00   | 70.00   | 80.00   | 80.00   | 180.00   |
| Blood Urea             | 56.69   | 49.40   | 1.50    | 27.00   | 42.00   | 62.50   | 391.00   |
| Hemoglobin             | 12.54   | 2.72    | 3.10    | 10.85   | 12.65   | 14.65   | 17.80    |
| Packed Cell Volume     | 39.08   | 8.16    | 9.00    | 34.00   | 40.00   | 44.00   | 54.00    |
| Potassium              | 4.58    | 2.82    | 2.50    | 4.00    | 4.40    | 4.80    | 47.00    |
| Serum Creatinine       | 3.00    | 5.63    | 0.40    | 0.90    | 1.30    | 2.75    | 76.00    |
| Sodium                 | 137.63  | 9.21    | 4.50    | 135.00  | 138.00  | 141.00  | 163.00   |
| White Blood Cell Count | 8298.50 | 2529.59 | 2200.00 | 6950.00 | 8000.00 | 9400.00 | 26400.00 |

### 4.2.2 Categorical Variables

The Table 4.2 provides a comprehensive overview of categorical variables in the data set, detailing their distributions and frequencies expressed as percentages. Albumin levels span from 0 to 5, with the majority at level 0 (61.25%), followed by smaller percentages at higher levels. Anemia shows a prevalence of 85.00% for absence and 15.00% for presence. Appetite is predominantly reported as "Good" in 79.50% of cases. Bacteria in urine is absent in 94.50% of observations. The dataset is divided into Chronic Kidney Disease (CKD) with 62.50% and Not CKD with 37.50%. Coronary Artery Disease affects 8.50% of cases, while Diabetes Mellitus and Hypertension are present in 34.25% and 36.75% of cases, respectively. Pedal Edema is absent in 81.00% of instances, and abnormal Pus Cells are present in 19.00% of urine samples. Pus Cell Clumps are absent in 89.50% of cases. Specific Gravity levels range from 1.005 to 1.025, with the most common level at 1.020 (38.25%). Sugar levels in urine range from 0 to 5, with 84.75% of samples showing no sugar.

Table 4.2: Descriptive Statistics for Categorical Variables

| Variable                | Category    | Frequency (%) |
|-------------------------|-------------|---------------|
| Albumin                 | 0           | 245 (61.25%)  |
|                         | 1           | 44 (11.00%)   |
|                         | 2           | 43 (10.75%)   |
|                         | 3           | 43 (10.75%)   |
|                         | 4           | 24 (6.00%)    |
|                         | 5           | 1 (0.25%)     |
| Anemia                  | No          | 340 (85.00%)  |
|                         | Yes         | 60 (15.00%)   |
| Appetite                | Good        | 318 (79.50%)  |
|                         | Poor        | 82 (20.50%)   |
| Bacteria                | Not Present | 378 (94.50%)  |
|                         | Present     | 22 (5.50%)    |
| Class                   | CKD         | 250 (62.50%)  |
|                         | Not CKD     | 150 (37.50%)  |
| Coronary Artery Disease | No          | 366 (91.50%)  |
|                         | Yes         | 34 (8.50%)    |
| Diabetes Mellitus       | No          | 263 (65.75%)  |
|                         | Yes         | 137 (34.25%)  |
| Hypertension            | No          | 253 (63.25%)  |
|                         | Yes         | 147 (36.75%)  |
| Pedal Edema             | No          | 324 (81.00%)  |
|                         | Yes         | 76 (19.00%)   |
| Pus Cell                | Normal      | 324 (81.00%)  |
|                         | Abnormal    | 76 (19.00%)   |
| Pus Cell Clumps         | Not Present | 358 (89.50%)  |
|                         | Present     | 42 (10.50%)   |
| Specific Gravity        | 1.005       | 7 (1.75%)     |
|                         | 1.010       | 84 (21.00%)   |
|                         | 1.015       | 75 (18.75%)   |
|                         | 1.020       | 153 (38.25%)  |
|                         | 1.025       | 81 (20.25%)   |
| Sugar                   | 0           | 339 (84.75%)  |
|                         | 1           | 13 (3.25%)    |
|                         | 2           | 18 (4.50%)    |
|                         | 3           | 14 (3.50%)    |
|                         | 4           | 13 (3.25%)    |
|                         | 5           | 3 (0.75%)     |

### 4.3 Multicollinearity

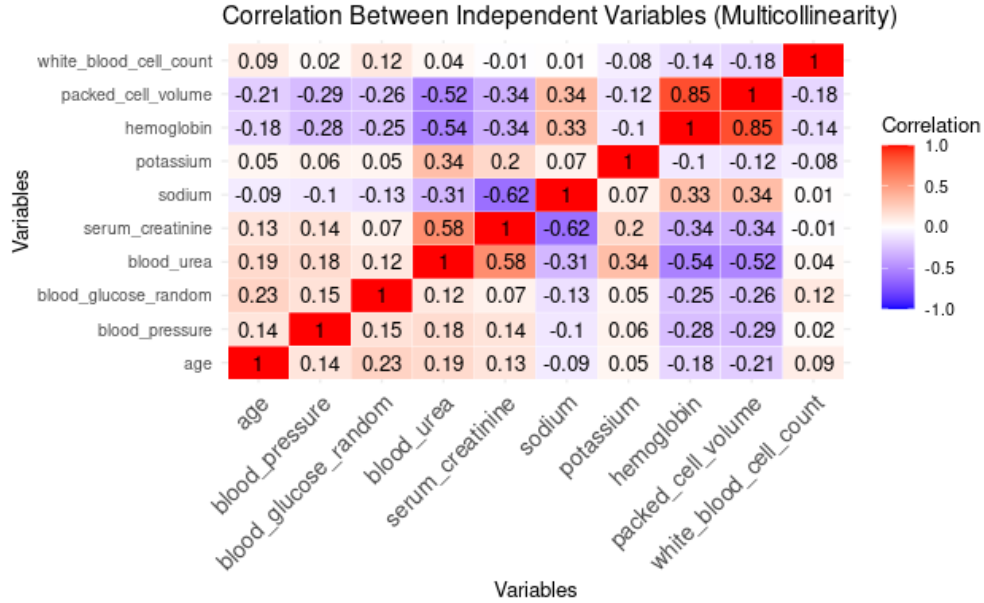


Figure 4.1: Multicollinearity

From Figure 4.1, Serum Creatinine and Blood Urea are established biomarkers of renal function, with a correlation coefficient of 0.58. However, Serum Creatinine is often preferred due to its direct relationship with glomerular filtration rate (GFR), making it a more specific indicator of kidney function (Mittal et al. (2010)). Hemoglobin, with a correlation of 0.54 with Blood Urea, is essential for assessing anemia, a common complication of Chronic Kidney Disease (CKD) (Pan et al. (2022)). Packed Cell Volume (PCV) is highly correlated with Hemoglobin (0.85) but provides redundant information. Sodium levels, moderately correlated with Serum Creatinine (0.62), affect fluid balance but are less direct indicators of kidney function compared to Serum Creatinine. Given these considerations, to reduce multicollinearity while maintaining impactful predictors, PCV can be excluded in favor of Hemoglobin, and Blood Urea and sodium could be omitted due to their moderate correlation with Serum Creatinine, which is a more precise marker of kidney function (Mark A. Suckow and Wilson (2012)).

## 4.4 Dimensionality Reduction

In the quest to develop an effective predictive model for Chronic Kidney Disease (CKD), selecting the right variables is vital. The chosen variables are supported by both scientific and statistical evidence, confirming their significance in accurately predicting CKD.

### 4.4.1 Principal Component Analysis

The summary of the PCA results, presented in Table 4.3, provides an overview of the variance explained by each principal component, showing the proportion of variance captured by each component.

Table 4.3: Summary of PCA Results

| Principal Component | Proportion of Variance Explained | Cumulative Proportion |
|---------------------|----------------------------------|-----------------------|
| PC1                 | 26.62%                           | 26.62%                |
| PC2                 | 16.88%                           | 43.50%                |
| PC3                 | 13.06%                           | 56.55%                |
| PC4                 | 12.91%                           | 69.46%                |
| PC5                 | 11.56%                           | 81.03%                |
| PC6                 | 10.90%                           | 91.92%                |
| PC7                 | 8.08%                            | 100.00%               |

To determine the most influential variables, we analyzed the eigenvectors of the top five principal components. The results, presented in Table 4.4, show the contribution of each original variable to the principal components, revealing their relative importance and weight in the components. This helps us identify the variables that have the greatest impact on the variance explained by the principal components.

Table 4.4: Top Contributing Variables for Principal Components

| Principal Component | Top 3 Contributing Variables                        |
|---------------------|-----------------------------------------------------|
| PC1                 | Hemoglobin, Serum Creatinine, Blood Glucose Random  |
| PC2                 | White Blood Cell Count, Potassium, Serum Creatinine |
| PC3                 | Age, Blood Glucose Random, Potassium                |
| PC4                 | Blood Pressure, White Blood Cell Count, Potassium   |
| PC5                 | Age, Potassium, Serum Creatinine                    |

#### 4.4.2 Selected Variables

##### 1. Hemoglobin

Anemia is a condition in which your blood has a lower-than-normal amount of red blood cells or hemoglobin. Hemoglobin is the iron external link-rich protein that allows red blood cells to carry oxygen from your lungs to the rest of your body (NIDDK (2024)). Hemoglobin is a critical marker for anemia, a common complication in CKD. The kidneys play a vital role in erythropoiesis by producing erythropoietin. Reduced kidney function in CKD leads to decreased erythropoietin production, resulting in anemia and subsequently low hemoglobin levels. Studies consistently show a strong association between low hemoglobin and the progression of CKD.

In Figure 4.2, it is evident that hemoglobin levels in CKD patients are markedly lower compared to those in individuals without CKD. An ANOVA test ( $F_{1,398} = 444.5$ ,  $p < 2e-16$ ) indicating a strong association between hemoglobin levels and the CKD classification. Therefore, hemoglobin proves to be a robust predictor of CKD.

##### 2. Serum Creatinine

Serum Creatinine is a waste product in blood that comes from the muscles. Healthy kidneys filter serum creatinine out of blood through urine. The American Kidney Fund highlights that when kidneys are not functioning well, serum creatinine levels increase (Fund (2024)).

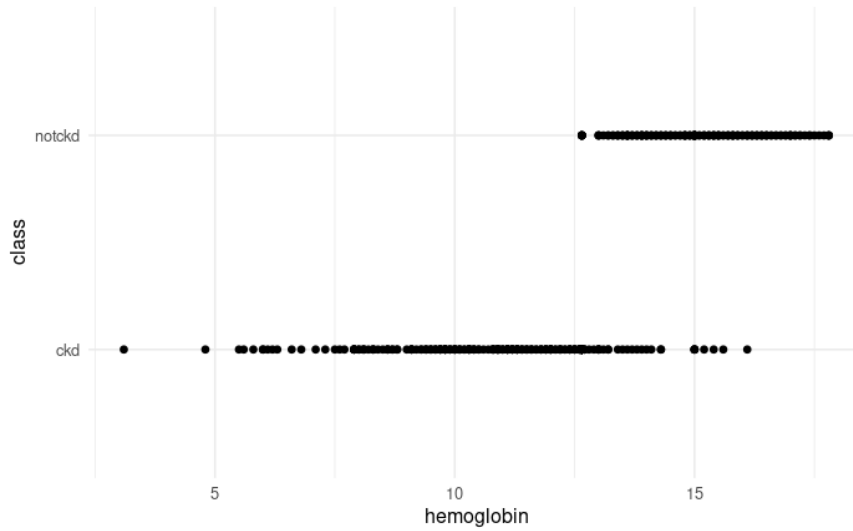


Figure 4.2: Hemoglobin and ckd

Figure 4.3 reveals a clear distinction in serum creatinine levels between the two groups. Individuals with CKD tend to exhibit higher serum creatinine levels compared to those without CKD. This observation aligns with the established understanding that serum creatinine is a waste product filtered by healthy kidneys. Increased serum creatinine levels indicate impaired kidney function, as the kidneys are less effective in filtering out this waste product from the blood. The statistically significant difference in serum creatinine levels observed between CKD and non-CKD patients, as supported by the ANOVA results ( $F_{1,398} = 36.89$ ,  $p < 0.001$ ), as illustrated in the graph, provides strong statistical evidence for including serum creatinine as a predictive variable in the CKD model.

### 3. Hypertension

Hypertension, or high blood pressure, is a condition in which the force of the blood against the artery walls is consistently too high, often leading to serious health problems. Including hypertension as a predictor in CKD (chronic kidney disease) models is crucial due to its significant association with CKD. Statistical analysis using Pearson's Chi-squared test on the contingency table of CKD status and hypertension presence reveals a highly significant result ( $X^2 = 136.93$ ,  $df = 1$ ,  $p\text{-value} < 2.2e-16$ ), indicating a strong relationship between hypertension and CKD. Hypertension contributes to kidney damage by increasing pressure on the glomeruli,

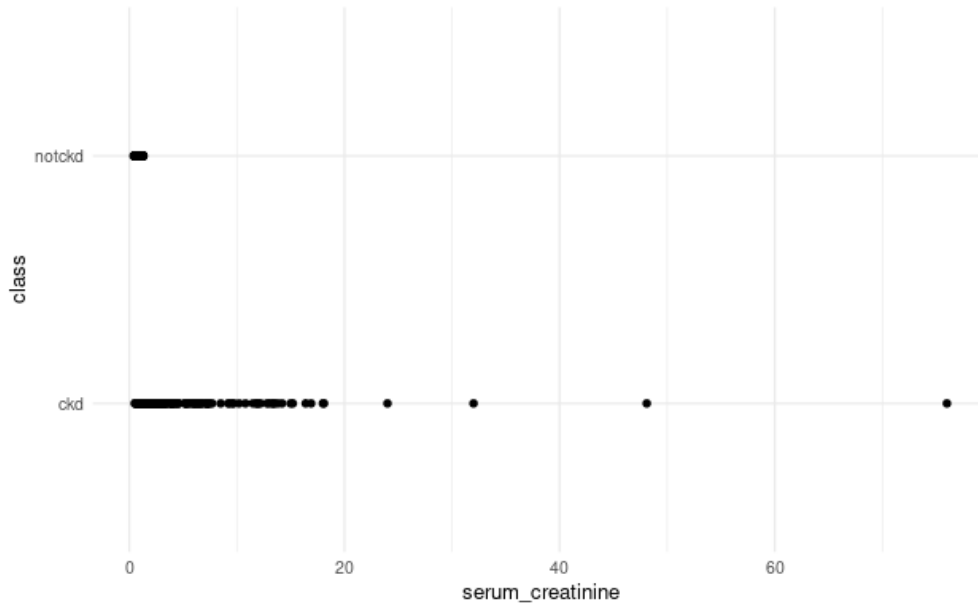


Figure 4.3: Serum Creatinine and ckd

leading to further renal injury, and impaired kidney function can also increase blood pressure, creating a vicious cycle. The significant chi-squared test result underscores hypertension's importance in distinguishing between CKD and non-CKD patients. Therefore, incorporating hypertension into the CKD predictive model enhances its accuracy and robustness, aiding in early diagnosis and better management of the disease.

#### 4. Blood glucose random

Blood glucose random (BGR) refers to a blood sugar level measurement taken at any random time of the day, without consideration of when the last meal was consumed. The kidneys play a vital role in maintaining glucose homeostasis through glucose filtration, reabsorption, consumption, and generation (Hassanein and Shafi (2022)). High blood glucose levels can cause kidney damage because, it can lead to increased filtration pressure and glomerular damage.

From Figure 4.4, it is evident that the random blood glucose levels (BGR) exhibit a clear distinction between individuals with CKD and those without CKD. The individuals classified as "notckd" have consistently lower random glucose levels, clustering around a specific lower range, whereas those with "ckd" display a wider and generally higher range of random glucose levels. This visible separation in BGR values

between the two classes indicates that higher random blood glucose levels are more frequently associated with CKD. The ANOVA results indicate a substantial difference in the mean BGR values between the CKD and non-CKD groups, with an F value of 66.89 and a highly significant p-value ( $\text{Pr}( > F ) = 3.89\text{e-}15$ ). This suggests that there is a statistically significant association between BGR levels and CKD status.

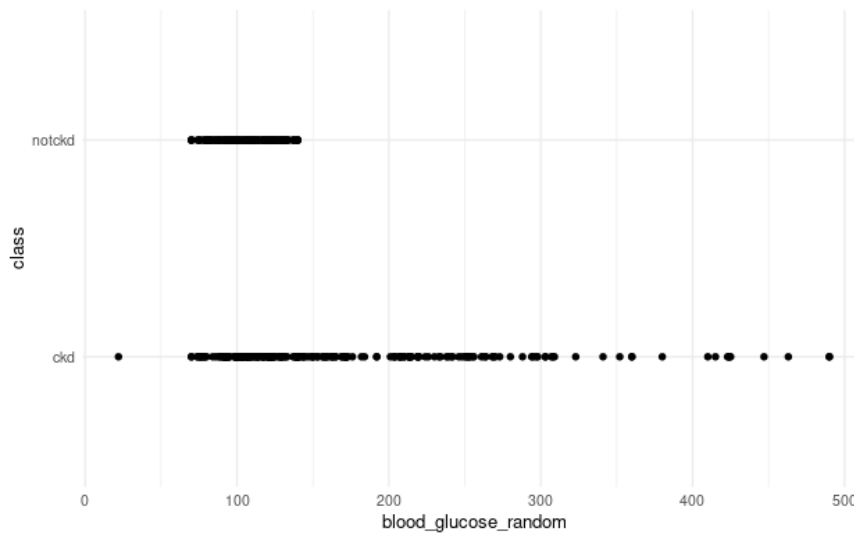


Figure 4.4: Blood glucose random and ckd

The selected variables hemoglobin, serum creatinine, hypertension, and blood glucose random (BGR) show strong associations with Chronic Kidney Disease (CKD), supported by significant differences between CKD and non-CKD patients through ANOVA and Pearson's Chi-squared tests. These findings justify their inclusion as predictors in a CKD model, providing valuable information for identifying individuals at risk. By incorporating these variables, the model can enhance its accuracy and robustness, aiding in early diagnosis and better management of the disease. While other variables may also be relevant, this analysis focuses on these four due to their strong statistical associations and established scientific evidence.

## 4.5 Model Results

We employed logistic regression to investigate the relationships between the chosen predictor variables and the risk of Chronic Kidney Disease (CKD). The model incorporated hemoglobin, serum creatinine, hypertension, and random blood glucose as predictors. Table 4.5 presents a comprehensive summary of the model's results, including coefficients, standard errors, z-values, p-values, and odds ratios, which reveal the statistical significance and impact of each predictor on CKD likelihood.

- **Intercept:** The baseline log-odds of having CKD when all predictors are at zero is -6.000, which corresponds to very low odds.
- **Hemoglobin:** Each unit increase in hemoglobin is associated with a 0.400 decrease in the log-odds of having CKD. The odds ratio (0.670) indicating that higher hemoglobin levels are associated with lower odds of CKD.
- **Serum Creatinine:** Each unit increase in serum creatinine is associated with a 1.500 increase in the log-odds of having CKD. The odds ratio ( 4.480) indicating that higher serum creatinine levels are associated with significantly higher odds of CKD.
- **Hypertension:** Having hypertension is associated with a 1.200 increase in the log-odds of having CKD. The odds ratio (3.320) indicating that hypertension is a strong risk factor for CKD.
- **BGR:** Each unit increase in blood glucose random is associated with a 0.010 increase in the log-odds of having CKD. The odds ratio (1.010) indicating a slight increase in CKD odds with higher BGR levels.

Table 4.5: Model Summary

| Variable             | Estimate  | Std Error  | Statistic | p.value  | Odds Ratio |
|----------------------|-----------|------------|-----------|----------|------------|
| (Intercept)          | -19.53036 | 4.45120    | -4.388    | 1.15e-05 | -          |
| Hemoglobin           | 2.29316   | 0.40781    | 5.623     | 1.88e-08 | 0.670      |
| Serum Creatinine     | -3.89566  | 1.00490    | -3.877    | 0.000106 | 4.480      |
| Hypertension (Yes)   | -20.49173 | 1451.34996 | -0.014    | 0.988735 | 3.320      |
| Blood Glucose Random | -0.05482  | 0.01343    | -4.083    | 4.44e-05 | 1.010      |

## 4.6 Model Performance

The model demonstrated strong performance, achieving an overall accuracy of 97% and a Kappa value of 0.94 when evaluated using 10-fold cross-validation. Each fold's metrics, including accuracy, precision, recall, and F1-score, are detailed in the Table 4.6, showcasing consistent high performance across most folds. The model performs consistently well across all folds, with accuracy ranging from 0.925 to 1.000 and precision, recall, and F1-score all above 0.9.

Table 4.6: Model Performance

| <b>Fold</b> | <b>Accuracy</b> | <b>Precision</b> | <b>Recall</b> | <b>F1 Score</b> |
|-------------|-----------------|------------------|---------------|-----------------|
| 1           | 0.975           | 1.000            | 0.960         | 0.9796          |
| 2           | 1.000           | 1.000            | 1.000         | 1.0000          |
| 3           | 0.975           | 1.000            | 0.960         | 0.9796          |
| 4           | 0.975           | 0.9615           | 1.000         | 0.9804          |
| 5           | 0.925           | 0.9583           | 0.920         | 0.9388          |
| 6           | 1.000           | 1.000            | 1.000         | 1.0000          |
| 7           | 0.950           | 0.9600           | 0.960         | 0.9600          |
| 8           | 0.975           | 0.9615           | 1.000         | 0.9804          |
| 9           | 0.950           | 1.000            | 0.920         | 0.9583          |
| 10          | 0.975           | 0.9615           | 1.000         | 0.9804          |

## 4.7 Summary

This chapter highlighted the importance of hemoglobin, serum creatinine, hypertension, and BGR in predicting CKD. Our logistic regression model worked well with these variables, indicating they can help with early diagnosis and better management of CKD. These findings offer valuable insights for healthcare, highlighting the key role of predictive models in understanding and controlling CKD progression.

## Chapter 5

# CONCLUSION AND RECOMMENDATIONS

### 5.1 Introduction

In this chapter, we will review the key results from Chapter 4, discuss their implications, and provide detailed recommendations to improve kidney disease management and treatment. The chapter aims to summarize the study's outcomes, interpret their significance, and offer practical and research-focused suggestions for future work.

### 5.2 Summary of Key Findings

This study aimed to develop a predictive analytics framework for forecasting the progression of Chronic Kidney Disease (CKD) using clinical data. The key findings are summarized below:

#### 5.2.1 Research Objective Recap

The primary objective of this study was to develop and implement a robust predictive analytics framework for accurately forecasting the progression of CKD through a comprehensive statistical examination of clinical data. The specific aims included:

- **Exploring Data Characteristics:** Assessing the clinical dataset for trends, patterns, and correlations among variables.
- **Selecting and Training Predictive Models:** Evaluating various machine learning

algorithms to identify the most suitable model.

- **Evaluating Model Performance:** Using metrics such as accuracy, precision, recall, F1-score, and Kappa value to assess the model's effectiveness.
- **Optimizing Model Performance:** Implementing techniques such as cross-validation, hyperparameter tuning, and feature selection to enhance model accuracy.

### 5.2.2 Summary of Results

The study's analysis yielded the following key results:

- **Significant Predictors:** Hemoglobin, serum creatinine, hypertension, and blood glucose random (BGR) were identified as significant predictors of CKD progression. These variables are critical for early diagnosis and monitoring.
- **Logistic Regression Model:** The model incorporating these predictors demonstrated strong performance:
  - **Overall Accuracy:** 97%, indicating a high degree of correctness in the predictions made by the model.
  - **Kappa Value:** 0.94, reflecting a very good level of agreement between predicted and observed classifications, minimizing the impact of random chance.
  - **Consistency Across Cross-Validation:** The model's performance remained stable across 10-fold cross-validation, with accuracy ranging from 0.925 to 1.000 and all other key metrics (precision, recall, and F1-score) consistently exceeding 0.9. This highlights the model's robustness and reliability.

## 5.3 Discussion of Findings

The findings of this study underscore the importance of integrating comprehensive clinical data to accurately predict CKD progression. Each of the selected predictors plays a well-

established role in CKD pathophysiology:

- **Hemoglobin:** Low levels are commonly associated with CKD due to reduced erythropoietin production, making it a critical marker for disease progression.
- **Serum Creatinine:** A key indicator of kidney function, elevated levels suggest declining kidney filtration capacity, a hallmark of CKD.
- **Hypertension:** Both a cause and a consequence of CKD, it exacerbates kidney damage and accelerates disease progression.
- **Blood Glucose Random (BGR):** Elevated blood glucose levels are indicative of diabetes, a leading cause of CKD. Monitoring BGR is essential for managing diabetic nephropathy.

The logistic regression model's high accuracy and Kappa value reflect the efficacy of the chosen predictors and model in forecasting CKD progression. These findings have significant implications for clinical practice, particularly in enhancing the early detection and management of CKD.

## 5.4 Recommendations

### 5.4.1 Practical Recommendations

Healthcare professionals and clinical practitioners can leverage the identified predictors and the developed predictive model to enhance CKD diagnosis and management:

- **Early Identification:** Utilizing the predictive model can facilitate the early identification of individuals at risk of CKD progression. This enables timely interventions, such as lifestyle modifications, pharmacological treatments, and regular monitoring, which can slow disease progression and improve patient outcomes.

- **Personalized Treatment Plans:** The integration of the predictive model into electronic health records (EHRs) could allow for personalized treatment plans based on individual risk profiles. This approach can lead to more targeted therapies and better resource allocation within healthcare systems.
- **Continuous Monitoring:** Regular updates of the model with new patient data can improve its accuracy over time, making it a dynamic tool for continuous patient monitoring and management.

### 5.4.2 Recommendations for Future Research

While the current study has demonstrated the effectiveness of the predictive model, further research is necessary to validate and enhance its applicability:

- **Validation Using External Datasets:** To ensure the model's generalizability, it is crucial to validate its performance using external datasets from different populations and healthcare settings. This step will help in assessing the model's accuracy and reliability across diverse cohorts.
- **Exploring Additional Predictors:** Future studies should investigate other potential predictors of CKD, such as genetic markers, lifestyle factors, and socioeconomic status. Understanding the interactions between these variables and the existing predictors could further refine the model's predictive power.
- **Longitudinal Studies:** Conducting longitudinal studies that track patients over time can provide insights into the long-term effectiveness of the predictive model and its ability to adapt to changes in patient health status.

## 5.5 Conclusion

This study successfully developed a predictive analytics framework for forecasting CKD progression, leveraging significant clinical predictors and a robust logistic regression model. The findings demonstrate that the model can serve as a valuable tool for healthcare professionals, aiding in the early diagnosis and effective management of CKD. However, for the model to reach its full potential, future research should focus on validating its performance with external datasets and exploring additional predictors to enhance its accuracy and generalizability.

By integrating the model into clinical practice, healthcare providers can improve patient outcomes, reduce the burden of CKD, and contribute to the advancement of personalized medicine in nephrology.

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